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Original article

Rapid beef quality detection using spectra pre-processing methods in electrical impedance spectroscopy and machine learning

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Summary The fraudulent practice of beef adulteration is a growing concern, as it violates consumer rights. Electrical impedance spectroscopy (EIS) combined with machine learning has emerged as a widely used approach to identify low-quality meat. Unlike traditional biochemical methods that require expensive instruments, complex sample preparation, and chemical reagents, EIS is a cost-effective alternative. However, EIS data are susceptible to temperature fluctuations, requiring a waiting period under consistent temperature conditions for data stabilisation before measurements. This process becomes impractical when dealing with a large number of samples. To overcome this limitation, standardisation, normalisation, and smoothing methods were introduced in the meat quality detection based on EIS data. A recognition model for detecting carrageenan adulteration in beef was established. Under an inconsistent temperature condition, by applying the spectra pre-processing methods to the prediction dataset, the model accuracy reached 84%, whereas the accuracy of the unprocessed prediction dataset dropped to 54%. This study demonstrates that acquiring EIS data under consistent temperature conditions is unnecessary if proper spectra pre-processing methods are applied. By eliminating the waiting time for data stabilisation, this practical approach enhances the efficiency and accuracy of meat quality detection.

Keywords Beef quality inspection, carrageenan adulteration, electrical impedance spectroscopy, machine learning, repeated freeze–thaw, spectra pre-processing, water adulteration.

Introduction

Beef consumption has increased significantly in China over the past two decades due to its high-nutritional value and consumer preference (Li *et al.*, 2018). However, ensuring beef quality and safety has become a major challenge for both producers and consumers due to various forms of food fraud involving adulteration or mislabeling of beef products (Rogberg-Munoz *et al.*, 2016; Li *et al.*, 2022). One common type of food fraud is injecting water into beef tissues to enhance their weight and appearance (Ballin, 2010). This not only deceives consumers economically but also poses health risks due to potential contamination by microorganisms or chemicals (Wheeler *et al.*, 1993). Another type of food fraud is adding carrageenan into meat products as a thickening agent to increase their water-holding capacity (Aga *et al.*, 2021). Carrageenan has been controversial for its possible negative effects on gastrointestinal health (Liu *et al.*, 2021a). A third type of food fraud is subjecting meat products to repeated freeze–thaw cycles (Gao *et al.*, 2020; Hassoun *et al.*, 2020). Subjecting meat products to multiple freezing and thawing processes can deteriorate their quality in various ways, such as odour, colour, and texture (Leygonie *et al.*, 2012; Mohammed *et al.*, 2021).

Electrical impedance spectroscopy (EIS) has been widely used for analysing biological objects in various fields, such as medical (Gamal *et al.*, 2018), agronomic (Liu *et al.*, 2021b), and food (Pliquett, 2010; Huang *et al.*, 2023). Unlike traditional biochemical methods that require expensive instruments, complex sample

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preparation, and chemical reagents, EIS is a cost-effective alternative. In medical field, EIS has proven to be valuable for diagnosing diseases, monitoring physiological changes (Gamal *et al*., 2018). In agronomic field, EIS has been used for measuring soil moisture (Liu *et al*., 2021b), plant growth (Repo *et al*., 2005; Cseresnyes *et al*., 2019), and fruit ripeness (Li *et al*., 2019; Ibba *et al*., 2020). In the field of food quality control, EIS can effectively assess the freshness (Perez-Esteve *et al*., 2014), water content (Yang *et al*., 2013; Wei *et al*., 2017), salt content (Ma *et al*., 2016; Chen *et al*., 2017; Schivazappa *et al*., 2017), and fat content (Afonso *et al*., 2020).

Among various food products, meat is one of the most important sources of protein and other nutrients for human consumption. Meat quality can be affected by many factors during production, processing, storage, and distribution. Therefore, it is essential to monitor and evaluate the quality of meat products using reliable and accurate methods. The EIS is one such method that can measure the electrical properties of meat tissues, which reflect their physical, chemical, and biological characteristics. Researchers have evaluated the quality of meat by EIS and machine learning including beef aging (Damez *et al*., 2008), the salt content in pork chops (Kaltenecker *et al*., 2013), storage condition evaluation in pork, the water adulteration in pork (Bai *et al*., 2018; Leng *et al*., 2019, 2020), evaluation of fish freshness (Sun *et al*., 2020), and the fat quality of beef (Afonso *et al*., 2020), etc. These studies have shown that EIS provides useful information on meat quality indicators, and classification models of meat quality can be established by combining with machine learning.

However, previous machine learning research based on EIS required obtaining both the training dataset and prediction dataset under consistent temperature conditions. Additionally, researchers had to wait until the sample temperature stabilised before performing measurements due to the high sensitivity of EIS data to temperature variations (Henningsson *et al*., 2005; Sekulic *et al*., 2015). Moreover, EIS data are also highly influenced by sample size, which requires cutting samples into similar sizes and shapes before the examination. These requirements pose limitations on the practical detection of meat quality using EIS, especially when dealing with large sample volumes. Standardisation, normalisation, and smoothing methods have been commonly used as spectra pre-processing techniques in infrared spectroscopy, nuclear magnetic resonance spectroscopy and Raman spectroscopy (Rinnan *et al*., 2009; Giraudeau *et al*., 2014; Huang *et al*., 2018). However, their applicability to EIS remains unexplored. Furthermore, no studies have compared the characteristics of EIS for carrageenan adulterated beef, water adulterated beef, and repeated freeze–thaw beef.

The work was conducted with the following specific objectives: (1) The discrimination of water adulterated beef, carrageenan adulterated beef, repeated freeze–thaw beef and fresh beef by EIS; (2) The establishment of a discrimination model based on Light Gradient Boosting Machine (LightGBM) for distinguishing between carrageenan-adulterated beef and fresh beef; (3) The application of spectra pre-processing methods to EIS, including standardisation, normalisation and smoothing techniques; and (4) The identification of the optimal smoothing method for the predication set.

Material and methods

Beef samples

The experimental sample used in this study was the yellow cattle tenderloin obtained from a local supermarket (RT-MART). The beef samples were stored in a refrigerator at a temperature of 5 °C and were fully wrapped in cling film until the start of the experiment. Previous studies have indicated the necessity of consistently maintaining the sample size of training and prediction sets and conducting experiments under the same temperature in order to obtain reliable experimental results (Damez *et al*., 2008; Bai *et al*., 2018; Huh *et al*., 2021). Consequently, the prediction set required for the established model need to be obtained under identical conditions, which reduces the model's generalisation ability. Moreover, maintaining consistent sample sizes poses operational challenges during high-volume detection. In order to overcome the limitation, we proposed to apply spectra pre-processing methods on prediction set. To test the robustness of the proposed method in the presence of different sample volumes and temperature fluctuations, we intentionally introduced inconsistencies in sample sizes between the training and prediction sets during the preparation process. For the training set, each beef sample was cut into cubes of uniform size of $3 \times 3 \times 3$ cm. On the other hand, for the prediction set, samples were cut into different sizes, ranging from $2 \times 2 \times 2$ cm to $5 \times 5 \times 5$ cm cubes. The average weight of beef samples was 40.5 g, with a maximum of 99.1 g and a minimum of 12.9 g.

Preparation of water adulterated beef

Tap water was injected into the beef samples using disposable sterile syringes, and the samples were massaged during the procedure to help them absorb the water. The percentage of injected water in each beef sample was: 0%, 5%, 10%, 15% and 20% (refer to Fig. S1 in the supplementary materials).

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Preparation of carrageenan adulterated beef
To prepare a carrageenan solution, food-grade carrageenan powder was mixed with distilled water at a ratio of 0.3–100. The carrageenan adulterated beef samples were prepared using similar methods as those mentioned in section 2.1.1. The injected carrageenan percentage in each beef sample was 0%, 5%, 10%, 15%, and 20% (refer to Fig. S2 in the supplementary materials). In order to develop a model for detecting carrageenan adulteration in beef, a total of 150 beef samples with 15% carrageenan were prepared.

Preparation of repeated freeze–thaw beef
The beef samples underwent freeze–thaw cycles for 1, 2, and 3 times (refer to Fig. S3 in the supplementary materials), with each cycle lasting 6 h at the refrigerated temperature of −18 °C. Before the measurement, the beef samples were thawed at a room temperature around 25 °C, and it took 6 h to completely thaw the beef.

Figures S1, S2, and S3 in the supplementary materials depict the colour changes observed in beef samples after adulteration and freeze–thaw cycles. The injection of water and carrageenan resulted in a noticeable loss of vibrant colours in the beef samples. However, it is important to highlight that these injected beef samples regained their original vivid colours after being refrigerated at 5 °C for 3 h. This phenomenon could potentially mislead consumers who rely solely on visual cues to assess the quality of beef.

EIS measurements

The experimental setup consisted of an Inductance, Capacitance and Resistance (LCR) meter (TH2816A, Tonghui, Changzhou, China), a self-made 3D printed test platform (with a square base and a sample holder with holes for the probes), and a cooling incubator (LRH-150; Bluepard, Shanghai, China), as shown in Fig. 1. The beef samples were penetrated to a depth of 1 cm by inserting two copper probes, spaced 1.5 cm apart.

Measurements were performed at 22 frequencies ranging from 50 Hz to 20 kHz (0.05, 0.06, 0.08, 0.1, 0.12, 0.14, 0.18, 0.24, 0.3, 0.36, 0.42, 0.5, 1, 2, 3, 5, 7, 9, 12, 15 and 20 kHz), the following parameters were recorded while maintaining a voltage of 1 V: the parallel equivalent capacitance (Cp), series equivalent capacitance (Cs), complex impedance (Z), phase angle (θ), reactance (X), and impedance (R). Since beef is electrically anisotropic (Dean *et al.*, 2008), its electrical properties depend on the angle between the electrical field direction and the muscle fibre direction. Therefore, the electrical current along the muscle fibres (white arrow in Fig. 1) was measured rather than across them (yellow arrow in Fig. 1). The EIS data of each sample were acquired for three times at different positions and averaged.

In this study, the training set consisted of 100 samples. The measurements for the training set were conducted in a temperature-controlled cooling incubator set at 8 °C, as shown in Fig. 1. Each sample was fully wrapped in cling film and placed in the incubator. Approximately 30 min were given for the sample to stabilise before the measurement of EIS data. This ensured a consistent temperature condition during the measurements for the training set.

On the other hand, the prediction set comprised of 50 samples. The measurements for the prediction set were carried out at room temperature without air conditioning, which means that the temperature was not controlled or regulated. Additionally, there was no waiting time for the data stabilisation before conducting the measurements for the prediction set. This resulted in an inconsistent temperature condition during the measurements for the prediction set.

EIS pre-processing methods

First, outliers were excluded using the 3σ rule (Suraci & Oliveira, 2019). Second, the data were standardised (Gal & Rubinfeld, 2019) to bring all predictors to the

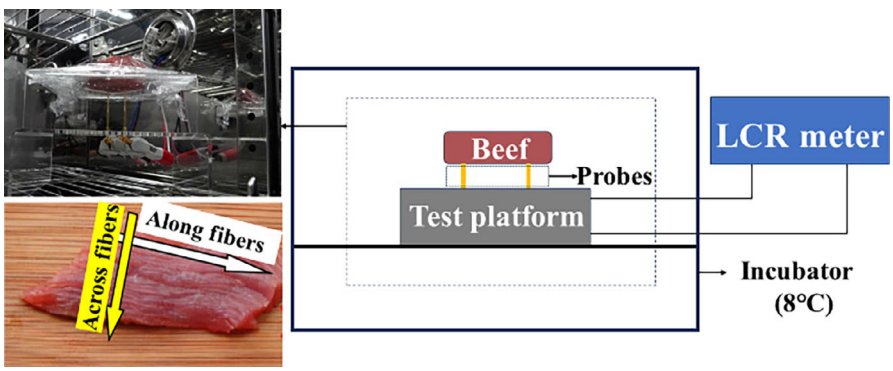

Figure 1 EIS measurement setup for model training.

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same scale. In order to mitigate the impact caused by variations in sample size on EIS data, the min–max normalisation (Jain *et al.*, 2018) was performed. This transformation aimed to standardise the range of predictors to a fixed scale of 0–1, thereby ensuring comparability of EIS data across different sample sizes.

In order to decrease the influence of the inconsistent temperature condition and eliminate abnormal data, including Gaussian smoothing (GS) (Wink & Roerdink, 2004), Savitzky–Golay smoothing (SG) (Press & Teukolsky, 1990), exponential weighted moving average (EWM) (Haynes *et al.*, 2012), weighted moving average (WMA) (Shih & Tsokos, 2008) and locally weighted scatterplot smoothing (LOWESS) (Bogusz *et al.*, 2014), were applied in the prediction dataset.

Machine learning

Light gradient boosting machine

The LightGBM (Mienye & Sun, 2022) is a powerful machine learning tool based on gradient boosting trees. It excels in capturing complex non-linear relationships and discovering underlying patterns within similar datasets. The LightGBM uses a leaf-wise growth strategy, which contributes to both high accuracy and efficiency during model training. By iteratively training decision trees and automatically selecting important features, LightGBM progressively enhances the model's fitting ability. This adaptive learning approach effectively uses feature information within similar datasets. Furthermore, LightGBM performs automatic feature selection during the training process, giving priority to features based on their importance. This ensures a more effective use of feature information within similar datasets.

Validation

In the binary classification model, various metrics were applied to evaluate the goodness-of-fit and performance of the model. These metrics included accuracy, precision, recall, and F1 score. The equations and definitions of above metrics are presented in Table S1 in supplementary materials. True positive (TP), true negative (TN), false positive (FP), and false negative (FN) are defined in a previous study (Chicco & Jurman, 2020).

Machine learning workflow

The workflow of machine learning-based carrageenan adulteration detection in beef is illustrated in Fig. 2. The study began by designing two different conditions of EIS measurements for the model training and prediction, respectively. Both datasets were pre-processed through normalisation and standardisation. Then five smoothing algorithms were applied to the prediction set to improve the prediction performance. The machine learning algorithm was evaluated by various metrics in Table 1, confusion matrix (Heydarian *et al.*, 2022), and receiver operating characteristics (ROC) curve (Fawcett, 2006). The code and hyperparameters used in the experiment are available in the supplementary materials.

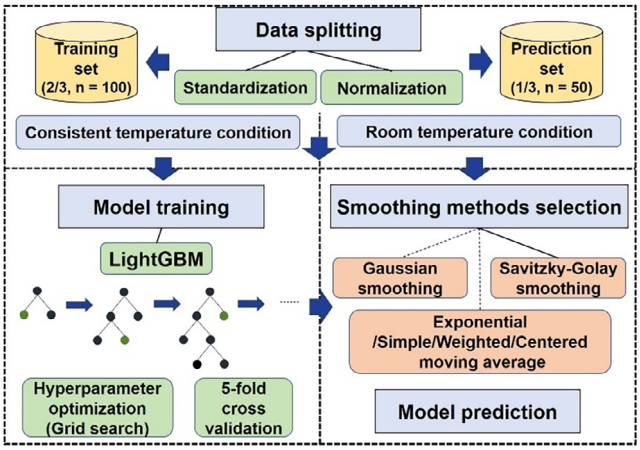

Figure 2 The workflow for adulterated beef detection based on LightGBM and spectra pre-processing.

Results and discussion

Electrical impedance spectroscopy analysis

Fig. 3a shows average EIS of the fresh beef and the low-quality beef. It can be found that the frequency-impedance curves had similar patterns for all samples. The impedance was high at low frequencies (below 1 kHz) due to the insulating effect of the cell membrane (Damez *et al.*, 2008), which prevented the current from flowing through the intracellular fluid. The cell membrane acted as a capacitor, allowing the current to pass through the cell at higher frequencies, resulting in lower impedance. Although the pattern was similar, it can be observed from Fig. 3a that the impedance of the lower quality beef was higher than that of the fresh beef, particularly in the case of the water adulterated beef, where the difference was significantly pronounced. Several factors, including drip loss, fibre structure damage, water exchange, and bacterial growth, may influence variations in impedance. These factors will be discussed subsequently.

Fig. 3b,c show the effect of different doses of carrageenan and water on impedance. This study fitted power functions to the frequency-impedance curves and found that the exponents decreased with increasing dose for both carrageenan and water. This means that the impedance decreased less rapidly with frequency as the dose increased. This could be explained

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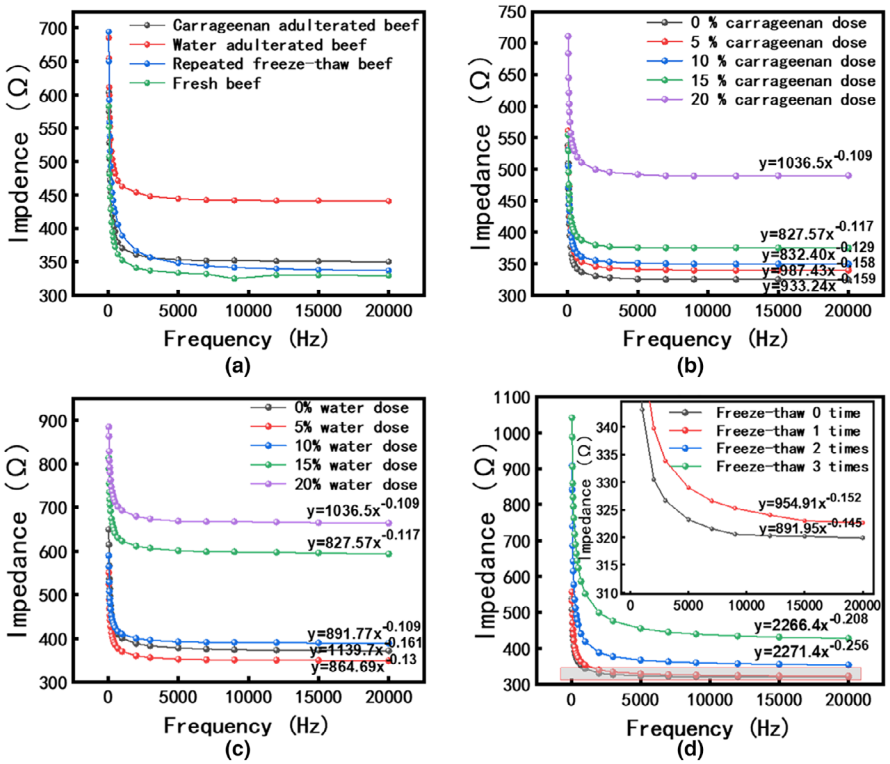

Figure 3 The impedance curves at various frequency. (a) Average impedance of beef samples. (b) Beef samples with different carrageenan doses. (c) Beef samples with different water doses. (d) Beef samples with different freeze–thaw cycles.

by the fact that carrageenan and water do not have the biological structure of beef tissue, such as cell membranes and cells, which affect the impedance at different frequencies. Therefore, injecting more carrageenan or water into beef reduces its impedance variation with frequency. It also can be noticed that the exponent for water decreased more sharply than that for carrageenan, which might be used as a criterion to distinguish between water adulterated and carrageenan adulterated beef.

Fig. 3d shows the effect of freeze–thaw cycles on impedance. In contrast to the other two types of lower quality beef, the exponent of frequency-impedance curves for the freeze–thaw beef increased with increasing number of cycles. This means that the impedance decreased more rapidly with frequency as the number of cycles increased. This could be caused by the damage to the cell membrane and the leakage of intracellular fluid due to repeated freezing and thawing (Guermazi *et al.*, 2014; Bai *et al.*, 2018). As a result, the current flowed more easily through the cells at high frequencies, and the impedance dropped more sharply.

The impedance data at 50 Hz and 12 kHz for different doses of carrageenan, water, and repeated freeze–thaw cycles were also analysed in this study. It can be observed that the impedance of the beef decreased first and then progressively increased with the correspondent increase in the amount of carrageenan or water at both low (Fig. 4a,b) and high (Fig. 4d,e) frequencies. This could be explained by two factors: (1) the addition of tap water and (2) the displacement of blood. Water is a conductive material that lowers the impedance of beef when injected with carrageenan or water (carrageenan contains water). However, when too much carrageenan or water is injected, it displaces the blood in the beef, which also contains conductive ions. Moreover, it disrupts the original structure and orientation of the beef fibres, which makes it more difficult for current to flow through. Other studied also confirmed our results (Leng *et al.*, 2019). Fig. 4c,f show that the impedance of beef samples increased with increasing number of freeze–thaw cycles at both low and high frequencies. However, some previous studies (Wei *et al.*, 2017; Dell'Osa *et al.*, 2021) reported that the rapid formation of ice crystals during freezing damages the fibres organisation of muscles, leading to altered insulating properties of cellular membranes. This resulted in increased ion mobility and low impedance. The opposite results could be

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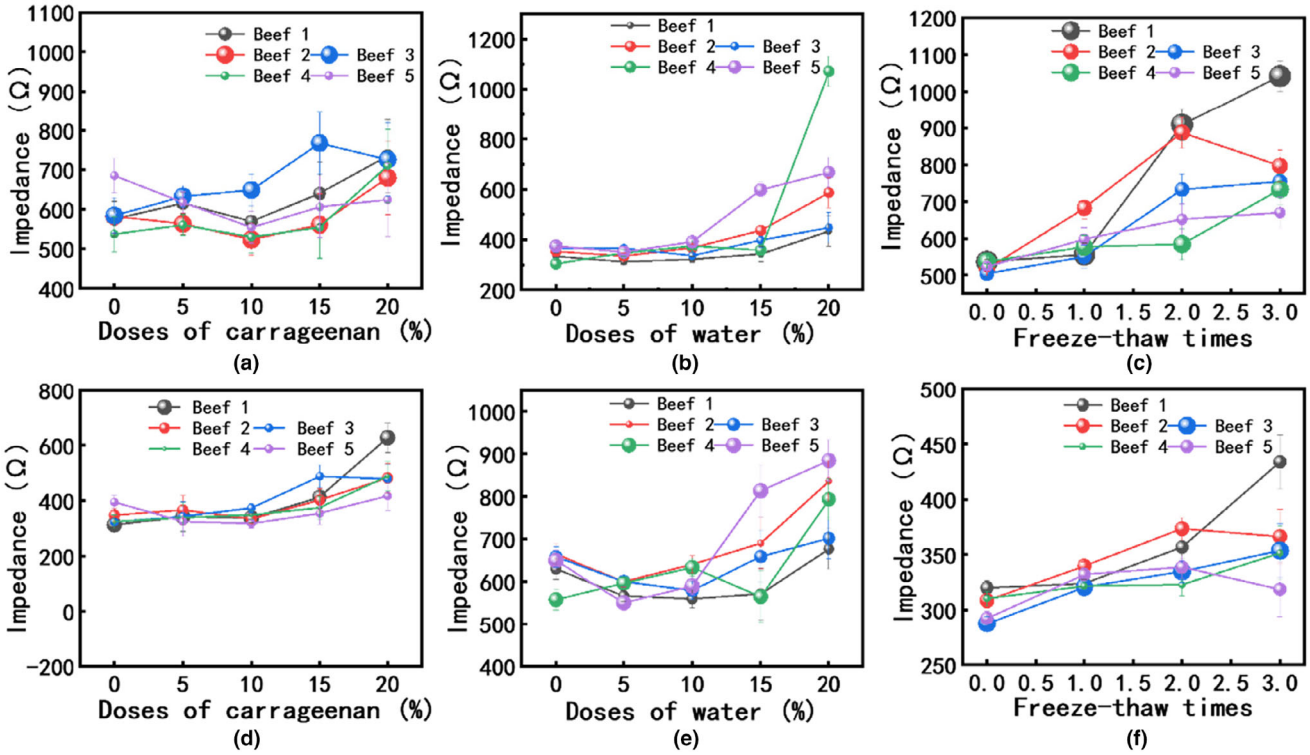

Figure 4 Effects on impedance with respect to different doses of adulterants and freeze–thaw cycles. (a) Carrageenan, (b) water and (c) freeze–thaw cycles at 50 Hz. (d) Carrageenan, (e) water and (f) freeze–thaw cycles at 12 kHz.

explained by two factors which may be more dominant in this experiment: (1) the growth of bacteria and (2) drip loss. Bacteria can grow in beef samples after repeated freezing and thawing, and they have lower conductivity than beef tissue (Pal *et al.*, 2016). Furthermore, repeated freezing can damage and break cells, causing ions to leak out with the blood (Abie *et al.*, 2021). This study removed the blood before testing, which also removed some of the ions, resulting in higher impedance.

Development of recognition model

To effectively detect carrageenan adulterated beef, a recognition model was established based on LightGBM algorithm. A total of 60 features (Cp, Cs, Z, θ, X, and R at 50, 80, 100, 500, 1000, 3000, 5000, 12 000, 15 000, and 20 000 Hz, respectively) from electrical properties were imported into the model. The k-fold (k = 5) cross-validation technique (Fushiki, 2011; Wong & Yeh, 2020) and grid search technique (Adnan *et al.*, 2022) were used to train the model and optimise its performance. To investigate the viability of standardisation and normalisation methods used for the EIS-based model, Table 1 presents a comparison between the training and prediction results using the original data and the pre-processed data. Both the original and the pre-processed data shared the similar cross-validation scores, primarily attributed to the stringent control of beef sample size in the training set, thereby avoiding EIS variation during measurements. Owing to the disparity in sample sizes and temperatures among the prediction set, the discriminate rate of prediction set was significantly lower compared to the training set. The application of standardisation and normalisation resulted in a significant enhancement in prediction accuracy, increasing it from 0.54 to 0.7, which represents an improvement of ~30%. This results potentially arises from the effectiveness of

Table 1 Investigation of the identification results based on standardisation and normalisation

Pre-processing methods	Cross-validation score (N = 100)	Prediction accuracy (N = 50)	Processing time (s)
Standardisation & normalisation	0.88	0.70	0.48
None	0.85	0.54	0.51

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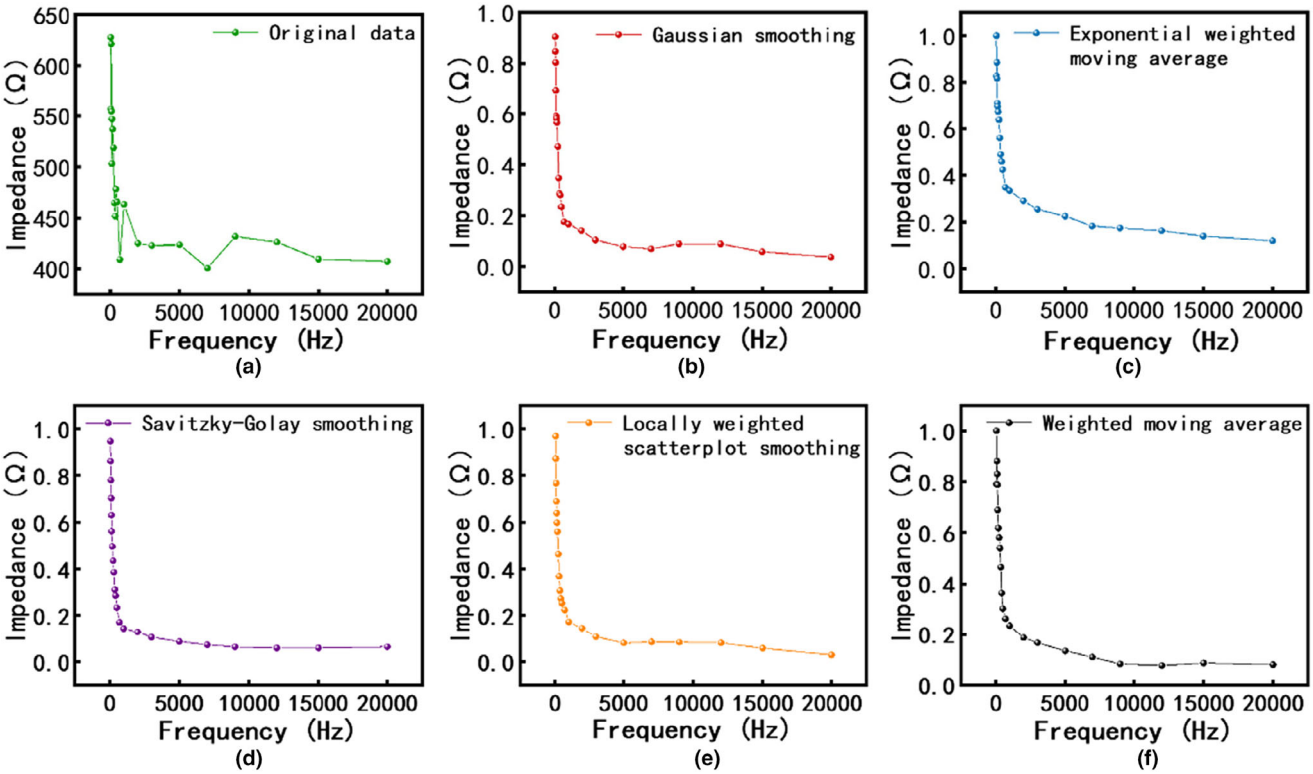

Figure 5 Comparison of impedance curves processed by different smoothing methods. (a) None. (b) Gaussian smoothing. (c) Exponential weighted moving average. (d) Savitzky–Golay smoothing. (e) Locally weighted scatterplot smoothing. (f) Weighted moving average.

normalisation and standardisation in mitigating the effects of feature scale, ensuring balanced contribution of each feature to model training, and appropriately scaling outliers to minimise their impact on the model.

To further optimise the performance of the model for achieving satisfactory results, this work performed five smoothing algorithms on the prediction data. Fig. 5 presents the comparison between the EIS curves of the original data and smoothed data. From Fig. 5 A, it could be observed that the impedance data obtained under the inconsistent condition exhibited significant fluctuations, which differed significantly from the curves collected under the consistent condition (Fig. 3). Furthermore, it was observed that most of the outliers resulting from the drastic temperature change of the sample in the high-frequency region were effectively eliminated by all five smoothing algorithms. However, only the SG method was able to effectively remove the outliers in the low-frequency region.

In general, the low-frequency region exhibited higher levels of noisy amplitudes than the high-frequency region, which posed challenges for certain smoothing methods in mitigating their impact effectively. Specifically, SG smoothing uses polynomial fitting to estimate the trend of a signal and applies a sliding window for smoothing. Compared to other smoothing algorithms, SG smoothing can better preserve the overall trend of signals in low-frequency regions, adapt to the local characteristics of low frequency signals, and introduce less bias (Press & Teukolsky, 1990). The detailed smoothing prediction results were compared and summarised in Table 2, where the model successfully discriminated between the fresh and the carrageenan adulterated beef with an accuracy range of 0.58–0.84. Based on the results, it can be concluded that the use of SG smoothing yielded the highest prediction accuracy of 0.84 (an improvement of 20% over unsmoothed data), precision of 1, recall of 0.68, and F1 score of 0.81.

To specifically assess the model using spectral preprocessing methods, confusion matrices and ROC curves were calculated in this study. Fig. 6a shows that the area under the ROC curves for the model using pre-processed data is 0.99, indicating its excellent discrimination capability. Fig. 6c,d display the confusion matrices, which depict the predicted results for the fresh beef (class 0) and carrageen adulterated beef (class 1). According to the results, raw data only correctly detected three out of the 25 fresh beef samples,

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Table 2 The discrimination results of the 5 smoothing methods for the prediction set

	Smoothing methods	Prediction accuracy	Precision	Recall	F1 score	Processing time (s)
Standardisation & normalisation	None	0.70	0.70	0.68	0.69	0.48
	GS	0.74	1.00	0.48	0.64	0.42
	EWM	0.64	0.64	0.64	0.64	0.46
	SG	0.84	1.00	0.68	0.81	0.45
	LOWESS	0.58	1.00	0.16	0.27	0.74
	WMA	0.62	0.61	0.64	0.62	0.43

EWM, exponential weighted moving average; GS, Gaussian smoothing; LOWESS, locally weighted scatterplot smoothing; SG, Savitzky–Golay smoothing; WMA, weighted moving average.

whereas all fresh beef samples were accurately identified using pre-processed data. The recognition accuracy of the fresh beef increased from 0.12 to 1.

Differentiating between the carrageenan adulterated beef, the freeze–thaw beef, and the fresh beef based solely on EIS proves to be challenging, as their frequency-impedance curves in Fig. 3a are closely distributed. The LightGBM excels discovering underlying patterns within similar datasets (Wang *et al.*, 2021; Peng *et al.*, 2023). Thirty adulterated beef samples containing 5%–20% carrageenan, 30 freeze–thaw beef samples undergone 1–3 cycles, and 30 fresh beef samples were studied based on EIS and LightGBM. A cross-validation score of 0.86 in Table S2 demonstrates the feasibility of LightGBM in identifying similar electrical properties of various types of low-quality beef. Specifically, the accuracy for identifying fresh beef is 0.8, while the accuracy for identifying carrageenan adulterated beef and freeze–thaw beef is 0.9. The detailed results are presented by the confusion matrix in Fig. S4 in the supplementary materials.

In 2021, a study (Sun *et al.*, 2021) used ultra-high-performance liquid chromatography–tandem mass spectrometry (UHPLC–MS/MS, Price $1 million) to determine carrageenan content in meat and poultry. They performed chromatographic separation on the samples

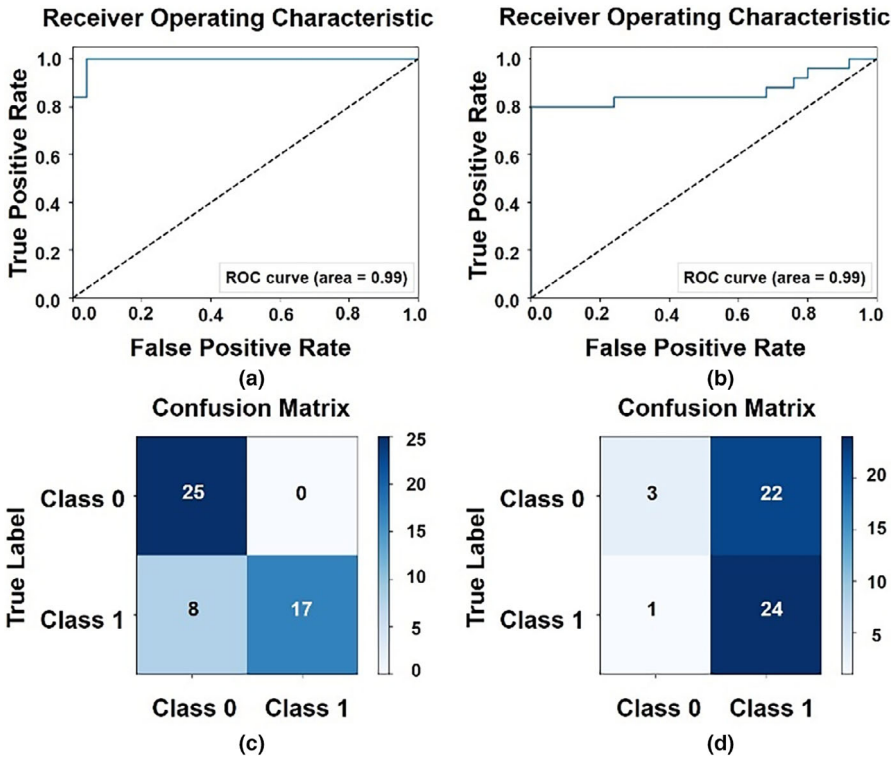

Figure 6 Performance comparison between the model on the pre-processed and raw data. (a) and (b) ROC curves. (c) and (d) confusion matrices. (Class 0 = fresh beef, Class 1 = carrageen adulterated beef).

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and used an electrospray ionisation, multiple reaction monitoring, and negative mode triple quadrupole mass spectrometer for quantitative analysis of carrageenan content. The expensive and intricate experiment procedures limit the applicability of their method to large-scale inspection. In contrast, our study aimed to detect the carrageenan adulterated, the water adulterated, and the freeze–thaw beef using a more affordable LCR instrument with a frequency range from 5 to 20 kHz (Price about $400). Moreover, we established a machine learning model to detect carrageenan adulteration in beef under inconsistent temperature conditions, achieving a respectable predicting accuracy of 0.84.

However, it is important to acknowledge some limitations in our work. The holding capacity, colour, moisture, and pH are the most important indices of meat quality and have been extensively studied (Cheng & Sun, 2008; Kim *et al*., 2014; Chumngoen *et al*., 2018; Tomasevic *et al*., 2021). They also found that the measurement of electrical properties can predict these quality metrics of meat (Byrne *et al*., 2000; Lee *et al*., 2000; Yang *et al*., 2013; Kim *et al*., 2014). Therefore, combining electrical property data with water holding capacity, colour, moisture, and pH allows for a comprehensive and accurate evaluation of beef quality and the development of a more precise predictive model. Unfortunately, conducting the necessary experiments was not feasible due to the lack of equipment for measuring water holding capacity and moisture. Furthermore, unethical practices often involve the use of various types of additives, such as gelatin, tara gum, cassia gum, and frequently, a combination of multiple additives. Regrettably, our study solely focused on developing a carrageenan discrimination model. Finally, maintaining a consistent temperature condition during model training remains essential for obtaining reliable results.

Conclusion

This study aims to investigate the application of EIS for rapid identification of low-quality beef. The findings demonstrate that the presence of carrageenan adulteration, water adulteration, and repeated freeze–thaw cycles have a significant impact on the bioimpedance measurements. This highlights the potential of EIS not only in distinguishing fresh meat from lower quality beef but also in identifying factors contributing to beef quality degradation. To overcome the limitations of EIS data acquisition, which requires a uniform size of samples and a consistent temperature environment, various spectra pre-processing methods have been applied on the EIS data for the first time. By comparing the results obtained from the raw data with the pre-processed data, which underwent standardisation, normalisation, and SG smoothing, the pre-processed data showed a 55% improvement in accuracy, reaching 0.84. These results exemplify the practicality and superiority of the developed method in real-world food regulation scenarios.

Considering the limitations of this study, further research will focus on the four aspects: (1) Developing detection models for different ratios and combinations of additives; (2) Evaluating the performance of the model under varying temperature conditions by applying spectral pre-processing methods on the training set; (3) Exploring additional pre-processing techniques for EIS, such as the first and second derivatives and standard normal variate; and (4) Investigating the holding capacity, colour, moisture, and pH of water adulterated and carrageenan adulterated beef.

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Conflict of Interest Statement

The authors declare no competing interests.

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All the materials in this manuscript are original.

Author contributions

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Peer review

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Data availability statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author.

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Data S1. Supporting information.

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